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University of New Brunswick unveils new supercomputer

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Virendra Bhavsar, centre, dean of the faculty of computer science at the University of New Brunswick; Gregory Kealey, left, UNB vice-president research; and John McLaughlin, right, university president and vice-chancellor, pose with UNB's new supercomputer in Fredericton.

The University of New Brunswick's newest supercomputer is a 1,818-kilogram behemoth that can crunch numbers 200 times faster than desktop computers.

And this computing powerhouse will be available not only to unviersity researchers, but also to those in the private sector, says the dean of the faculty of computer science at UNB.

"If [private sector researchers] want access, they'll have to do collaborative research with UNB so we can offer our facilities to them," says Virendra Bhavsar, dean of the faculty of computer science.

UNB researchers have worked with private sector researchers in the past using the university's older supercomputers, Bhavsar said.

"Some of the professors have been doing some work which is with other companies or with some of the other research institutions," he said.

The university showed off its new supercomputer on Thursday.

The Sun Microsystems unit has 192 processors and 50,000 gigabytes of storage space.

The unit, along with its 909-kg power supply and 1,362-kg air conditioning system, consumes 30 kilowatts of electricity.

"It is for research in many areas," said Bhavsar. "Humanities, science, computer science, engineering, anywhere people want to solve complex problems that would take a lot of time on desktop machines."

The supercomputer will help researchers in a number of ways, from making engineering models more accurate and realistic to helping discover the most effective radiation treatment for cancerous tumours.

The UNB supercomputer will gradually replace an older model acquired in 2004.

The new machine is part of a network of supercomputers at universities across Atlantic Canada.

The other universities participating in the Atlantic Computational Excellence Network (ACEnet) infrastructure are: Memorial University of Newfoundland (MUN), St. Mary's University and St. Francis Xavier University.

The supercomputer at UNB will be complemented by a state of the art videoconferencing centre, which will allow researchers to co-ordinate their efforts without having to travel.

The ACEnet infrastructure allows researchers to tap into more than one supercomputer at a time if additional processing power is required, he said.

Bhavsar said he expects there will be greater interest in UNB's supercomputers as more private sector researchers become aware of the available resources.

"I expect that as industry becomes more and more aware of this they will use it," he said.

"Some of these machines, people aren't aware they are in the province."